

## Problem 4

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**Problem 1** Determine the following limits. Use L'Hôpital's Rule if applicable.

(a)  $\lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 + 1}}$

**Explanation.** This limit is of the form:  $\frac{\infty}{\infty}$ . L.R. is applicable, but we don't need to apply it.

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 + 1}} &= \lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 \left(1 + \frac{1}{x^2}\right)}} \\ &= \lim_{x \rightarrow \infty} \frac{x}{|x| \sqrt{1 + \frac{1}{x^2}}} \\ &= \lim_{x \rightarrow \infty} \frac{x}{x \sqrt{1 + \frac{1}{x^2}}} \\ &= \lim_{x \rightarrow \infty} \frac{1}{\sqrt{1 + \frac{1}{x^2}}} \\ &= \frac{1}{\sqrt{1 + 0}} = 1 \end{aligned}$$

(b)  $\lim_{x \rightarrow -\infty} x^2 e^x$

**Explanation.** This limit is of the form:  $\infty \cdot 0$

$$\begin{aligned} \lim_{x \rightarrow -\infty} x^2 e^x &= \lim_{x \rightarrow -\infty} \frac{x^2}{e^{-x}} \left( \text{of the form } \frac{\infty}{\infty} \right) \\ &\stackrel{L.R.}{=} \lim_{x \rightarrow -\infty} \frac{2x}{-e^{-x}} \left( \text{of the form } \frac{\infty}{\infty} \right) \\ &\stackrel{L.R.}{=} \lim_{x \rightarrow -\infty} \frac{2}{e^{-x}} \\ &= 0 \end{aligned}$$

where "L.R." above an equals sign means that that equality is due to "L'Hôpital's Rule".

(c)  $\lim_{x \rightarrow \infty} x^{\frac{1}{x}}$

**Explanation.** This limit is of the form:  $\infty^0$

$$\begin{aligned}\lim_{x \rightarrow \infty} x^{\frac{1}{x}} &= \lim_{x \rightarrow \infty} e^{\ln\left(x^{\frac{1}{x}}\right)} \\ &= \lim_{x \rightarrow \infty} e^{\frac{1}{x} \ln x} \\ &= e^{\lim_{x \rightarrow \infty} \frac{\ln x}{x}} \quad \left(\text{limit is of the form } \frac{\infty}{\infty}\right) \\ &\stackrel{L.R.}{=} e^{\lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1}} \\ &= e^{\lim_{x \rightarrow \infty} \frac{1}{x}} \\ &= e^0 = 1\end{aligned}$$

(d)  $\lim_{x \rightarrow \infty} \left(1 + \frac{2}{x}\right)^x$

**Explanation.** This limit is of the form:  $1^\infty$

$$\begin{aligned}\lim_{x \rightarrow \infty} \left(1 + \frac{2}{x}\right)^x &= \lim_{x \rightarrow \infty} e^{\ln\left(\left(1 + \frac{2}{x}\right)^x\right)} \\ &= \lim_{x \rightarrow \infty} e^{x \ln\left(1 + \frac{2}{x}\right)} \\ &= e^{\lim_{x \rightarrow \infty} \frac{\ln\left(1 + \frac{2}{x}\right)}{1/x}} \quad \left(\text{limit is of the form } \frac{0}{0}\right) \\ &\stackrel{L.R.}{=} e^{\lim_{x \rightarrow \infty} \frac{-2x^{-2} \cdot \frac{1}{(1+2/x)}}{-x^{-2}}} \\ &= e^{\lim_{x \rightarrow \infty} \left(2 \left(\frac{1}{1 + \frac{2}{x}}\right)\right)} \\ &= e^2\end{aligned}$$

(e)  $\lim_{\theta \rightarrow 0^+} (\sin(\theta))^{\tan(\theta)}$

**Explanation.** This limit is of the form:  $0^0$

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$$\begin{aligned}
 \lim_{\theta \rightarrow 0^+} (\sin(\theta))^{\tan \theta} &= \lim_{\theta \rightarrow 0^+} e^{\ln((\sin(\theta))^{\tan(\theta)})} \\
 &= \lim_{\theta \rightarrow 0^+} e^{\tan(\theta) \ln(\sin(\theta))} \\
 &= \lim_{\theta \rightarrow 0^+} \frac{\ln(\sin(\theta))}{\cot(\theta)} \quad \left( \text{limit is of the form } \frac{\infty}{\infty} \right) \\
 &\stackrel{L.R.}{=} \lim_{\theta \rightarrow 0^+} \frac{\cos(\theta) \cdot \frac{1}{\sin(\theta)}}{-\csc^2(\theta)} \\
 &= \lim_{\theta \rightarrow 0^+} \left( \frac{\cos(\theta)}{\sin(\theta)} \cdot \frac{-\sin^2(\theta)}{1} \right) \\
 &= \lim_{\theta \rightarrow 0^+} (-\cos(\theta) \cdot \sin(\theta)) \\
 &= e^0 = 1
 \end{aligned}$$